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Controllable in situ surface restructuring of Cu catalysts and remarkable enhancement of their catalytic activity

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ABSTRACT: Supported metal catalysts play a pivotal role in the production of fuels and chemicals, and environmental remediation. However, identification and tailoring of the active metal sites on these catalysts still remain a formidable challenge, primarily due to their dynamic nature in catalytic reactions. Here, we report a simple and effective strategy for tuning supported Cu nanoparticles via controllable in situ surface restructuring to form coordinatively more unsaturated Cu sites, which exhibit a fourfold increase in catalytic turnover rate and excellent stability probed by ethanol dehydrogenation reaction. In situ measurements by transmission electron microscopy and X-ray absorption spectroscopy reveal intense phase transformation and consequent dynamic restructuring of Cu nanoparticles, induced by oxidation with temporarily added nitrous oxide and subsequent reduction under reaction conditions. Such restructuring is precisely controlled at the Cu surfaces without altering Cu dispersion and oxidation state, etc., enabling us to unambiguously identify the active sites during catalysis. This strategy, alternative to many sophisticated synthesis methods, has shown to be effective for engineering different Cu catalysts and in turn markedly enhancing their catalytic activity for a wide range of reactions varying from alcohol dehydrogenation to methanol steam reforming.

KEYWORDS: *Supported copper catalysts, active metal site, surface restructuring, in situ characterization, activity enhancement.*

1. INTRODUCTION

Metal catalysts, frequently dispersed in the form of metal nanoparticles on metal oxide and carbon supports, are widely used to increase rate and selectivity of a wide variety of chemical reactions in industrial processes, indispensable for production of fuels and chemicals as well as for pollution control and environmental remediation¹⁻³. Their catalytic properties often depend sensitively on the surface structures of the metal particles that can be tuned by tailoring the particle size, facet, morphology and underlying support¹⁻⁵. As such, numerous concepts and methods have been developed for controllable synthesis of metal nanoparticles, towards their superior catalytic performance and better understanding on the structure-property relationships⁶⁻⁸. Notwithstanding these efforts and advances, the well-synthesized metal particles are frequently subjected to structural change, preferentially on their surfaces, in response to reactive environments under pretreatment and reaction conditions⁹⁻²¹. These restructuring phenomena render it more complicated to gain the fundamental insights into the catalytic active sites and in turn to rationally design those sites^{12, 17, 22}. However, in many cases, such restructured metal particles exhibit significantly enhanced

catalytic activity or selectivity. For example, Vesborg et al. found that Cu nanoparticles on ZnO after treatment with intermediate mixtures of H₂ and CO exhibit a more flat shape with larger surface area and fraction of high-index surface facets, and thus enhanced methanol synthesis activity²³. Hutchings and coworkers reported that oxidation-reduction-oxidation treatment of Pd-Sn/TiO₂ catalysts induces the encapsulation of Pd-rich nanoparticles by amorphous SnO_x layers and consequently a much higher selectivity toward H₂O₂ synthesis from O₂ and H₂ reaction²⁴. Eren et al. found that induced by CO adsorption, the inert Cu (111) surface undergoes restructuring into nanoclusters and becomes highly active for water decomposition, a key step in water-gas shift reaction²⁵. These results demonstrate the structural dynamics of the metal catalysts and the catalytic consequences¹¹.

Moreover, these effects point out the feasibility of the controlled restructuring as a practical strategy to design of metal catalysts with reaction-desired structures, for example, via tuning restructuring environments ideally in situ under reaction conditions. However, developing such controllable methods still remains substantial challenges, mainly due to the lack of the in-depth understanding on

the dynamic structures of metal catalysts and the difficulty in tailoring such specific structures. In general, different structural parameters (e.g. crystallite size, shape, and oxidation state) for the metal particles tend to concurrently change and mutually interfere, as pervasively reported in the literatures²⁶⁻²⁷.

To address these problems, in this work, we report a facile method for the surface restructuring of Cu catalysts, induced by their intense phase transformation, involving N₂O oxidation treatment under the reaction conditions, and the consequently remarkable enhancement in their catalytic activity, as exemplified by Cu/SiO₂ catalyst in CH₃CH₂OH dehydrogenation to acetaldehyde (CH₃CHO), a probe reaction with well-established mechanism²⁸⁻²⁹. The reason for choosing Cu/SiO₂ as a model catalyst is that Cu catalysts are widely used in the industrially important reactions, such as methanol synthesis³⁰⁻³³, methanol stream reforming³⁴, and alcohol dehydrogenation³⁵. But it is known that the size and shape for Cu catalysts are more difficult to be controlled relative to noble metal catalysts because of their susceptibility to oxidation³⁶⁻³⁷.

We herein characterize the structures of the Cu nanoparticles on Cu/SiO₂ by complementary techniques including in situ transmission electron microscopy (TEM), X-ray diffraction (XRD), dissociative N₂O adsorption, X-ray photoelectron spectroscopy (XPS), and X-ray absorption spectroscopy (XAS). We observe that the restructuring, by careful choice of the treatment parameters, can be precisely controlled solely at the surfaces of the Cu nanoparticles without altering their other structural parameters (e.g. size, dispersion and oxidation state). In particular, it is noteworthy that the characterization of in situ aberration-corrected TEM, equipped with an advanced in situ gas cell consisting of gas-flow channel, electron-transparent windows and heating spiral³⁸⁻⁴⁰, enables us to visualize the dynamic structure of the Cu nanoparticles with atomic resolution at atmospheric pressure (~100 kPa), which bridges the pressure gap frequently encountered with the conventional environmental TEM operated below several hundred Pa⁴¹. Together with our reaction kinetic study and density functional theory (DFT) calculation, such controllably restructured Cu catalysts make it possible to unambiguously identify the active sites and understand the surface structure-reactivity correlation, a pervasive issue prerequisite for rational design of different metal catalysts^{12, 42}.

2. EXPERIMENTAL SECTION

2.1. Catalyst preparation. Cu/SiO₂ catalyst with 30.0 wt% CuO loading was prepared by a precipitation-gel method⁴³. Briefly, an aqueous solution of KOH (4 mol/L) was dropped into a solution of Cu(NO₃)₂ (1 mol/L) under vigorous stirring to form precipitates (until pH > 11). Next, a calculated amount of aqueous colloidal silica solution (40.0 wt%) was added to the above precipitates to form a gel, which was then aged at 373 K for 4 h under continuous stirring in the mother liquor. Finally, the slurry of the gel was filtered and thoroughly washed with distilled water, dried at 393 K overnight, and calcined at 723 K for 2 h in an air

flow. Cu/SiO₂-IM and Cu/ZnO-CP catalysts were prepared by incipient wetness impregnation and coprecipitation methods, respectively, and the detailed procedures were described in Supporting Information.

2.2. Catalyst performance and kinetic studies. Ethanol dehydrogenation reaction was carried out in a fixed-bed quartz microreactor at 473 K and atmospheric pressure. The catalysts (60-80 mesh) were diluted with quartz powders (60-80 mesh), and the reactants were introduced into the reactor by bubbling N₂ through a glass saturator filled with liquid ethanol. The sample was treated in situ under the reaction conditions, and the whole treatment process included three stages: Stage 1-before treatment (473 K, 5 kPa ethanol/N₂), Stage 2-during treatment (473 K, 5 kPa ethanol + 5 kPa N₂O/N₂), and Stage 3-after treatment (473 K, 5 kPa ethanol/N₂).

Kinetic isotope effects on ethanol dehydrogenation were examined by comparing the activities of deuterated ethanol (CH₃CH₂OD, CH₃CD₂OH and CD₃CH₂OH) with undeuterated ethanol under the same conditions as described above. CH₃CH₂OD was purchased from Sigma-Aldrich, and CH₃CD₂OH and CD₃CH₂OH from Cambridge Isotope Laboratories, Inc.

For examining the effect of treatment on Cu catalysts in other reactions, dehydrogenation of other alcohols (1-propanol, 2-propanol and 2-butanol), steam reforming of methanol, and hydrogenation of acetone, were carried out in the similar way as described above.

All stream lines from the reactor to the detector were kept above 393 K to avoid condensation of reactants and products. Reactants and products were analyzed by on-line gas chromatography (Shimadzu 2010 GC) using 5A molecular sieve column and Porapak Q column and TCD detector for N₂, O₂, CO, and CO₂ while AT-LZP-930 capillary column and FID detector for organic components. Conversions and selectivities were reported on a carbon basis, and rates as moles of reactants converted per mole of surface Cu sites per hour (h⁻¹), i.e. turnover frequencies (TOFs). The evolution of the gas-phase composition (H₂ and H₂O) during N₂O treatment was also monitored by mass spectrometry (Hiden HPR20).

2.3. Catalyst characterization. In situ TEM images and videos of Cu/SiO₂ under reaction conditions were taken on an aberration-corrected TEM (Titan, 80-300, FEI Company), operated at 300 KV with a gas cell holder system (Climate S3+, DENSolutions Company). The Cu/SiO₂ sample was dispersed on the window membranes supported across a heating chip for in situ TEM observations at reaction temperatures. The cell holder was connected to a gas supply system for the in situ sample cell with continuous flow at atmospheric pressure (H₂ at Stage 1-before treatment, 10 kPa H₂ + 5 kPa N₂O/Ar at Stage 2-during treatment, and H₂ at Stage 3-after treatment).

In situ XAS measurements for Cu/SiO₂ at the Cu K-edge were performed on the beamline 1W1B at the Beijing Synchrotron Radiation Facility. 20 mg sample was diluted with boron nitride, ground into fine powders, and then loaded in a custom flow cell inside a stainless steel chamber with

beryllium windows. The sample was treated in situ under the reaction conditions (see 2.2). All XAS spectra were collected at 473 K in the transmission mode during the treatment. XANES spectra were measured in the quick scanning mode (110 s per scan), and spectra for Cu/SiO₂ before treatment were taken and used as the reference spectrum.

The XAS spectra were analyzed using the program Athena software package. The background correction and normalization were carried out by fitting a linear polynomial to the pre-edge region and square polynomial to the post-edge region of the absorption spectrum. The E_0 value was determined by the maximum in the first derivative in the edge region. To estimate the relative concentrations of different Cu species in Cu/SiO₂ during treatment, XAS spectra of Cu, Cu₂O and CuO standard samples (representing the Cu⁰, Cu⁺ and Cu²⁺ species, respectively) were measured. Then, the in situ XAS spectra for Cu/SiO₂ were decomposed into a linear combination of the three Cu standards by the linear combination fitting (LCF) method with the IFEFFIT software package⁴⁴⁻⁴⁵ in the energy region from -20 to 60 eV. Fourier transformations were performed on k²-weighted EXAFS oscillations in the range of 2.7 – 12.6 Å⁻¹, employing Gaussian windows. For fitting the first coordination shell, the phase shifts and backscattering magnitudes were derived from a theoretical calculation from a standard sample using the FEFF6 code.

Quasi-in situ XPS measurements were performed on Axis Ultra spectrometer (Kratos, UK), equipped with monochromatic Mg K_α (1253.6 eV) radiation at a source power of 150 W, and also an in situ cell that allows heating and introduction of gases. All the binding energies were referenced to the contaminated C 1s peak at 284.8 eV. To trace the changes of the binding energy of the catalyst, 50 mg catalyst sample was loaded into the cell and treated under the reaction conditions as described above (see 2.2). After treatment, the sample was flushed by pure N₂, evacuated to 10⁻³ Pa, and then loaded into the measurement cell using the robot arm. The whole process was strictly controlled under anaerobic conditions. For comparison, the Cu LMM spectra for Cu and Cu₂O references were also measured. Cu was obtained by reducing CuO powders (Alfa Aesar) under 30 mL/min 20 kPa H₂/N₂ at 623 K for 2 h inside the in situ cell, and Cu₂O was supplied by Alfa Aesar without further treatment.

Other characterization experiments, including ex situ TEM, in situ XRD, and dissociative N₂O adsorption measurements, were described in the Supporting Information.

2.4. DFT Calculations. Theoretical treatments of the elementary steps involved in dehydrogenation of ethanol on Cu were performed using the plane-wave-based periodic DFT as implemented in the Vienna ab initio simulation package (VASP).⁴⁶⁻⁴⁷ All calculations were carried out within the generalized gradient approximation in the Perdew-Burke-Ernzerhof exchange-correlation functional (GGA-PBE) and projector augmented wave (PAW) pseudo-potentials with an energy cutoff of 396 eV.⁴⁸⁻⁵⁰ Convergence criteria were set up to 1×10⁻⁵ eV for energies and 0.05 eV Å⁻¹ for forces. The Monkhorst-Pack sampling method

was used to generate k-meshes for integration of the first Brillouin zone (6×6×6 for bulk structures and 6×6×1 for surfaces). The DFT-D₃ method with Becke-Jonson damping was used to account for the vdW interactions.⁵¹ All transition states were determined by the climbing image nudged elastic band method (CI-NEB) and dimer methods.⁵²⁻⁵³ The stretching frequencies were analyzed in order to determine every optimized transition structure has only one imaginary frequency.

The Cu(111) and (211) surfaces were modeled using a 4-layer slab respectively with p(3×3) and p(2×3) unit cells. For all the surface calculations, the top two layers were relaxed and the bottom two layers were fixed. The adsorption energy, E_{ads} , was calculated by $E_{\text{ads}} = E_{\text{adsorbate/Cu}} - E_{\text{Cu}} - E_{\text{adsorbate}}$, where $E_{\text{adsorbate/Cu}}$ is the total optimized energy of the adsorbates on Cu surface, E_{Cu} is the Cu surface energy, and $E_{\text{adsorbate}}$ is the optimized energy of the adsorbates in gas phase. Therefore, the more negative the adsorption energy, the stronger the adsorption. The activation energy, E_a , was calculated according to $E_a = E_{\text{TS}} - E_{\text{IS}}$, where E_{TS} and E_{IS} are the energy of the transition state and the initial state, respectively.

3. RESULTS AND DISCUSSION

3.1. N₂O-involved treatment effect on Cu/SiO₂ activity. Chosen as a model catalyst, Cu/SiO₂ with 30.0 wt% CuO was prepared by a precipitation-gel method⁴³. It was reduced in a H₂/N₂ flow at 623 K (defined as “Fresh” catalyst), and then exposed to 5 kPa CH₃CH₂OH in a N₂ flow until the CH₃CH₂OH dehydrogenation reaction reached its steady state at 473 K (defined as “Stage 1-before treatment”). Afterwards, 5 kPa N₂O was temporarily introduced to the reaction stream for 0.25 h (defined as “Stage 2-during treatment”). After N₂O was switched off, the catalyst was only exposed to the CH₃CH₂OH/N₂ stream again (defined as “Stage 3-after treatment”). Under all conditions in this study, CH₃CHO selectivity was higher than 95% before and after treatment, so we mainly focus on the evolution of catalytic activities (expressed as molar CH₃CH₂OH conversion rate per mole of surface Cu sites per hour, i.e. turnover frequency, TOF), which can be more apparently reflected by the ratios of the TOF values of Cu catalyst before and after treatment (denoted as TOF_{After}/TOF_{Before}).

Figure 1 shows the evolution of TOF and CH₃CHO selectivity on Cu/SiO₂ during its different stages at 473 K. Over the reaction period of “Stage 1-before treatment”, Cu/SiO₂ was active and selective for CH₃CH₂OH dehydrogenation to CH₃CHO, showing a steady-state TOF of 15.0 h⁻¹ and CH₃CHO selectivity of 98.3%. After the reaction for 3.4 h, 5 kPa N₂O was introduced into the CH₃CH₂OH/N₂ stream. Upon treatment for 0.25 h (N₂O was switched off at 3.7 h), the TOF value dramatically increased by about four times to 60.2 h⁻¹ (at 7.7 h) while the CH₃CHO selectivity was still as high as 98.7%. Such treated Cu/SiO₂ catalyst remained stable and did not show substantial decline in its activity over 70 h on-stream (~56 h⁻¹). Moreover, product analysis by on-line mass spectrometry (Figure S1) shows that at the steady-state of the CH₃CH₂OH dehydrogenation reaction, the H₂ signal (m/e=2) on Cu/SiO₂ after treatment was also

four times stronger than that before treatment by subtracting the background, in good agreement with the observed change of the TOF values for the $\text{CH}_3\text{CH}_2\text{OH}$ conversion. At the steady state, the intensity of H_2O signal ($m/e=18$) was always around its background level without obvious change before and after treatment. This result, together with the constant rates over 70 h after N_2O was completely flushed away from the reaction stream (Figure 1), excludes the possible involvement of any N_2O -derived oxygen species in the $\text{CH}_3\text{CH}_2\text{OH}$ dehydrogenation reaction on Cu/SiO_2 .

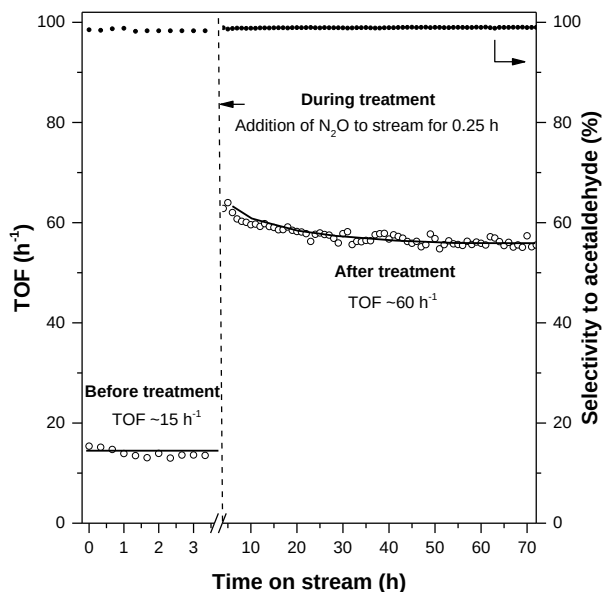


Figure 1. Evolution of ethanol dehydrogenation rates (TOFs) for Cu/SiO_2 catalyst at its different stage with time-on-stream. Reaction conditions: 30 mL/min, 5 kPa ethanol, N_2 balance, 473 K. Ethanol conversion before treatment was $\sim 3.0\%$; 5 kPa N_2O was introduced at 3.4 h and flowed for 0.25 h during treatment of Cu/SiO_2 catalyst.

3.2. N_2O -involved treatment effect on Cu/SiO_2 surface structure. Such dramatic enhancement of the Cu/SiO_2 activity is closely related to the structural change of the Cu surfaces after treatment, as revealed by the in situ aberration-corrected TEM images (Figure 2). The images were taken at 473 K under the reaction-relevant conditions (~ 100 kPa pressure), in which H_2 was used, instead of $\text{CH}_3\text{CH}_2\text{OH}$, with N_2O in the in situ TEM cell to eliminate the possible interference of $\text{CH}_3\text{CH}_2\text{OH}$ -derived carbon deposits on the Cu surfaces, considering the essentially identical treatment effect discussed later using $\text{CH}_3\text{CH}_2\text{OH}$ and H_2 . Clearly, upon N_2O treatment, the Cu nanoparticles of Cu/SiO_2 experienced a drastic transformation from a more faceted shape (Figures 2a, b) to a more spherical shape (Figures 2c, d) without changing their face-centered cubic structure. The surfaces became curved with more steps (Figure 2c), in comparison with the terminated flat $\{111\}$ and $\{100\}$ planes before treatment (Figures 2a, b). Moreover, defects (e.g. twin boundaries) on some Cu nanoparticles (Figure 2d) were formed upon the treatment, and the surface steps consisted of small facets like $\{111\}$, $\{100\}$

and $\{220\}$. These results reveal that the treated Cu surfaces become atomically rougher with more defect sites.

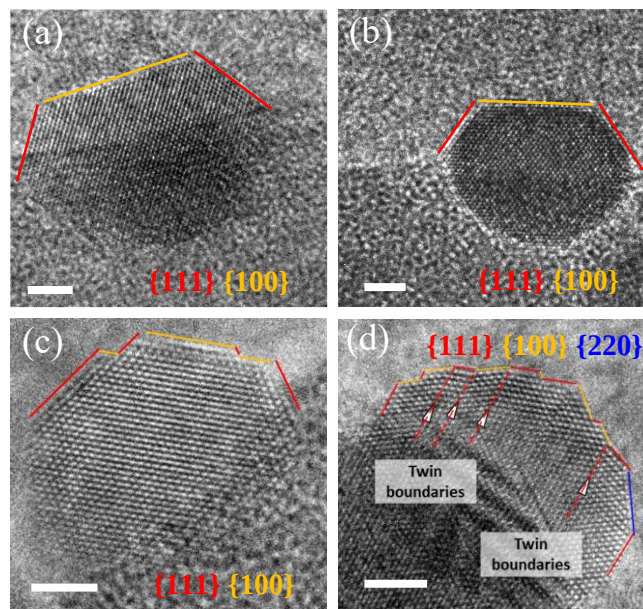


Figure 2. (a) and (b) In situ TEM images for Cu/SiO_2 before treatment (at ~ 100 kPa H_2 and 473 K). (c) and (d) In situ TEM images for Cu/SiO_2 after treatment (~ 100 kPa H_2 and 473 K). Scale bar for all TEM images: 2 nm; $\{100\}$, $\{111\}$ and $\{220\}$ represent the corresponding facets of Cu.

Complementary characterization results on the bulk and surface structures of Cu/SiO_2 before and after treatment were shown in Figures 3, S2 and S3, and Tables S2 and S3. The ex situ TEM images (Figures 3a, b) show that the Cu nanoparticles before and after treatment were distributed uniformly on the SiO_2 support with similar average diameters (5.5 nm vs. 5.9 nm) and narrow size distributions. Consistent with the TEM observation, in situ XRD (Figure S2) and dissociative N_2O adsorption (Table S2) results reveal no change in the Cu particle size (5.2–5.4 nm) and dispersion (20.2–24.1%) for the Cu/SiO_2 catalyst at its different stages (i.e. fresh, before treatment and after treatment). These results were further confirmed by the in situ extended X-ray absorption fine structure (EXAFS) spectra (Figure 3c) and the refined parameters for Cu/SiO_2 , including the Cu-Cu coordination number, Cu-Cu bond length and Debye-Waller factor (σ^2), remained intact after treatment (Table S3). The oxidation state of the Cu nanoparticles was characterized by XPS and Auger electron spectroscopy. As shown in Figure 3d, the signals at 932.3 eV and 918.6 eV respectively in the Cu 2p and Cu LMM spectra indicated the dominant presence of metallic Cu^0 on Cu/SiO_2 before and after treatment⁵⁴, which is consistent with the observed peaks at 8980.8 and 8993.5 eV, characteristic of metallic Cu^0 species⁴⁵, in the Cu K-edge X-ray absorption near edge structure (XANES) spectra, irrespective of treatment (Figure S3). Taken together, these complimentary results demonstrate that the average size, surface dispersion and metallic state of the Cu nanoparticles on Cu/SiO_2 retained essentially identical before and after treatment. Such identicalness enables us to unequivocally correlate

the observed enhancement in the activity of the Cu/SiO₂ catalyst after treatment solely to the restructuring and roughening of the Cu nanoparticle surfaces.

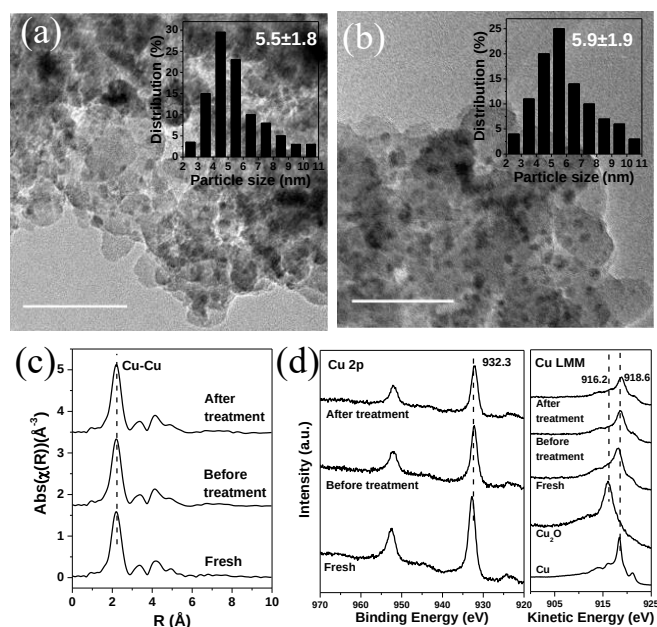


Figure 3. Ex situ TEM micrograph and size distribution for Cu/SiO₂ catalysts before treatment (a) and after treatment (b), respectively (scale bar: 50 nm). (c) Radial distribution functions for Cu/SiO₂ catalyst at different stages, including fresh, before treatment and after treatment. (d) Quasi-in situ XPS spectra of Cu 2p and Cu LMM signals for Cu/SiO₂ catalyst at different stages, including fresh, before treatment and after treatment.

3.3. Mechanistic understanding on enhanced Cu catalytic activity. To mechanistically understand the enhanced activity of the restructured Cu/SiO₂ catalyst, we carried out the reaction kinetic studies and DFT calculations. As shown in Table 1, within the kinetic-controlled regime (<10% conversion) at 473 K, the dehydrogenation rates for undeuterated and three deuterated ethanol molecules (i.e. CD₃CH₂OH, CH₃CD₂OH and CH₃CH₂OD) show that the normal kinetic isotope effect (KIE, defined as k_H/k_D) was greater than 1, irrespective of the N₂O treatment, only for CH₃CD₂OH (2.5 and 2.6) on Cu/SiO₂ before and after treatment, while the KIE values for CD₃CH₂OH (1.1 and 1.3) and CH₃CH₂OD (1.2 and 1.3) were very close to 1. These KIE results confirm that the rate-determining step for CH₃CH₂OH dehydrogenation on the Cu catalyst involves its β-C-H bond cleavage, independent of the N₂O treatment.

Table 1. Ethanol dehydrogenation rates and kinetic isotopic effects (KIEs) for Cu/SiO₂ catalysts before and after treatment^a

Stages	Reactants	TOF (h ⁻¹)	KIE(k_H/k_D)
Before treatment	CH ₃ CH ₂ OH	15.0	-
Before treatment	CH ₃ CH ₂ OD	12.7	1.2
Before treatment	CD ₃ CH ₂ OH	13.6	1.1
Before treatment	CH ₃ CD ₂ OH	6.1	2.5
After treatment	CH ₃ CH ₂ OH	60.2	-
After treatment	CH ₃ CH ₂ OD	47.4	1.3
After treatment	CD ₃ CH ₂ OH	46.4	1.3
After treatment	CH ₃ CD ₂ OH	23.1	2.6

^a Reaction conditions: 30 mL/min, 5 kPa ethanol, balance N₂, 473 K, ethanol conversions before treatment ~3%, ethanol conversions after treatment ~10%, acetaldehyde selectivity > 95%.

These KIE results, together with the CH₃CH₂OH pressure effect on the reaction rates (Figure S4), confirm that CH₃CH₂OH dehydrogenation on the Cu catalyst, independent of the N₂O treatment, follows the generally accepted mechanism on metal surfaces, which involves quasi-equilibrated dissociation of CH₃CH₂OH to form adsorbed CH₃CH₂O* intermediate and subsequent rate-determining β-hydride abstraction from CH₃CH₂O* to ultimately form CH₃CHO²⁸⁻²⁹.

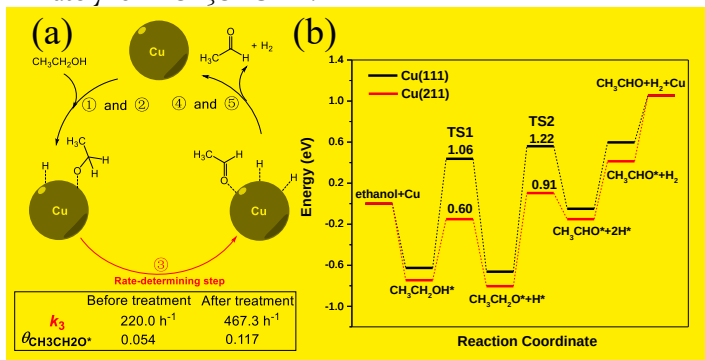


Figure 4. (a) Proposed reaction mechanism for ethanol dehydrogenation on Cu surfaces. (b) Gibbs free energy diagram obtained from DFT calculations for ethanol dehydrogenation on Cu (111) (Black) and Cu (211) (Red). All energies are relative to an ethanol molecule in gas phase and a clean Cu surface. Intermediates adsorbed on Cu surfaces are marked with a star.

Based on these kinetic and isotopic results, we proposed a sequence of elementary steps (Figure 4a and Table S1), and derived the CH₃CH₂OH dehydrogenation rate equation (Eq. S16). Clearly, the estimated coverage of the CH₃CH₂O* intermediate ($\theta_{CH_3CH_2O^*}$) and rate constant for the rate-determining step (k_3) (Figure 4a, Figures S4-6, Table 1 and Table S4) reveal that the N₂O treatment led to a significant increase in both the adsorption of the CH₃CH₂O* intermediate ($\theta_{CH_3CH_2O^*}$: 0.117 vs. 0.054) and activity for the C-H bond cleavage (467.3 h⁻¹ vs. 220.0 h⁻¹) on the restructured Cu surfaces. This mechanistic insight into the promoting effect of the N₂O treatment is in good agreement with the general conclusion on the adsorption and

catalytic properties of the metal catalysts with coordinately more unsaturated sites on the atomically rough surfaces^{34,55}. This conclusion is further corroborated by the DFT calculation result (Figures 4b and S7), showing that compared with the flat surfaces (e.g. Cu (111)), the stepped surfaces (e.g. Cu (211)) increased the adsorption strength of the $\text{CH}_3\text{CH}_2\text{O}^*$ intermediate (-0.80 eV on Cu (211) vs. -0.66 eV on Cu (111)) and decreased the reaction free energy barriers of the kinetic-relevant C-H bond cleavage step (0.91 eV on Cu (211) vs. 1.22 eV on Cu (111)).

3.4. Origin of Cu surface restructuring induced by N_2O -involved treatment. The controlled surface restructuring of the Cu/SiO₂ catalyst is driven by oxidation of the Cu nanoparticles with N_2O under the $\text{CH}_3\text{CH}_2\text{OH}$ dehydrogenation conditions, as evidenced from their dynamically structural evolution visualized by the in situ, time-resolved XANES spectra (Figure 5) and TEM images (Figure S8, Movies 1-4). Figure 5a shows the Cu K-edge XANES spectra for Cu/SiO₂ collected over a period of ~ 1.0 h during N_2O addition to the reaction stream at 473 K. The two signals at 8980.8 eV and 8993.5 eV observed for Cu/SiO₂ before treatment signifies the dominant presence of metallic Cu⁰ species⁴⁵. The two signals, upon introduction of N_2O , shifted to 8982.1 eV and 8995.9 eV, respectively, indicating a rapid oxidation of Cu⁰ to Cu ^{δ^+} ($1 < \delta < 2$), by referring to the characteristic signals of Cu⁺ (at 8981.1 eV and 8994.4 eV) and Cu²⁺ (at 8984.8 eV and 8996.2 eV)⁴⁵. They then slowly shifted to 8981.0 eV and 8993.9 eV, indicative of partial reduction of the Cu ^{δ^+} species by $\text{CH}_3\text{CH}_2\text{OH}$ during treatment. When N_2O was switched off, the signals ultimately shifted back to 8980.9 eV and 8993.6 eV, reflecting the presence of pure Cu⁰ species for Cu/SiO₂ after treatment. Corresponding quantitative analysis for such evolution of the Cu oxidation states (Figure 5b) shows that the fractions of the Cu species on Cu/SiO₂ varied from 100% Cu⁰ at 3.4 h before treatment rapidly to a mixture of 37% Cu⁰, 43% Cu⁺ and 20% Cu²⁺ at 3.5 h and of 82% Cu⁰, 14% Cu⁺ and 4% Cu²⁺ at 3.7 h during treatment, and ultimately back to >99% Cu⁰ (with <1% Cu⁺) at 4.0 h after treatment. Considering the accuracy of the fitting method⁵⁶, these data demonstrate that the Cu species on Cu/SiO₂ underwent intense redox reactions during treatment and were then completely reduced to their original metallic state after treatment, in line with the findings by in situ XRD and XPS (Figures S2 and 3d).

From the in situ TEM images for the individual Cu nanoparticles on Cu/SiO₂ under the reaction conditions (Figure S8 and Movies 1-4), it is noted that for the fresh Cu/SiO₂ catalyst reduced by H₂ at 623 K, the metallic Cu nanoparticles were stable at 473 K in the sample cell filled with H₂ (Figure S8a and Movie 1), which excludes the possible effects of electron beam irradiation on the dynamic behaviors of the Cu species discussed in this work. When N_2O was introduced, the Cu nanoparticles were oxidized rapidly to form Cu/CuO_x core-shell nanostructures (Figure S8b and Movie 2) with a metallic Cu core (~ 7 nm diameter) dispersed with a CuO_x layer (~ 4 nm in thickness), in accordance with the aforementioned in situ-XANES results

(37% Cu⁰ + 63% CuO_x) during treatment (Figure 5b). After N_2O was switched off, the CuO_x layer was reduced, and the core-shell structure immediately disappeared to consolidate into a spherical particle with a ~ 10 nm diameter within 24 s (Figure S8c). Such significant shape change of the Cu nanoparticles, especially with the appearance of defects (e.g. twin boundaries, Figure 2d), can be mainly attributed to their intense phase transformation and severe contraction once N_2O was switched off under the reaction conditions (Figure S8c and Movie 3). Afterwards, the restructured Cu nanoparticles remain unaltered in shape even after prolonged annealing in H₂ (Figure S8d and Movie 4). This stability reflects the irreversibility of the observed structural change for the Cu nanoparticles under the reaction conditions, which is largely different from the frequently reported reversible reconstruction induced for example by adsorbates on the metal surfaces⁵⁷⁻⁵⁹.

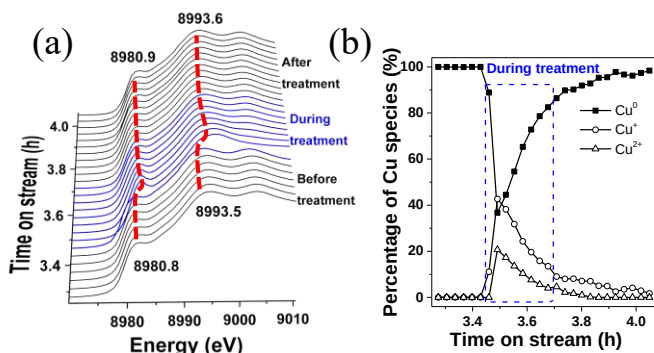


Figure 5. (a) In situ, time-resolved XANES spectra for Cu/SiO₂ catalyst collected at different stages with time-on-stream. (b) Percentage of Cu species of different oxidation state on Cu/SiO₂ estimated from XANES spectra with time-on-stream.

Such oxidation-induced restructuring process is further verified by the effects of treatment gases and temperatures. As shown in Table 2, when N_2O (Entry 1) was replaced by a stronger oxidant of O₂ (Entry 2), similar enhancement was observed for Cu/SiO₂ ($\text{TOF}_{\text{After}}/\text{TOF}_{\text{Before}} = 4.1$), however, the Cu nanoparticle size after O₂ treatment increased from 5.3 nm to 9.6 nm (Figures S2 and S9). Differently, when H₂O (a weaker oxidant, Entry 3), Ar (an inert gas, Entry 4), and H₂ (a reducing gas, Entry 5), instead of N_2O , were individually introduced to the ethanol dehydrogenation feed (5 kPa $\text{CH}_3\text{CH}_2\text{OH}$, N_2 balance) for 0.25 h, $\text{TOF}_{\text{After}}/\text{TOF}_{\text{Before}}$ was 1.2, 1.0, and 1.0, respectively, close to unity. While keeping N_2O unchanged, replacing $\text{CH}_3\text{CH}_2\text{OH}$ by H₂ (Entry 6) and even inert Ar (Entry 7) to treat the Cu/SiO₂ catalyst led to a similar activity, and $\text{TOF}_{\text{After}}/\text{TOF}_{\text{Before}}$ reached 3.6 and 3.7, respectively. Furthermore, the effect of treatment temperatures (Entries 8-10) shows higher $\text{TOF}_{\text{After}}/\text{TOF}_{\text{Before}}$ values (2.9 - 4.8) at higher temperatures in the range of 453 - 493 K, apparently due to the facilitated oxidation of the Cu nanoparticles with N_2O . These results corroborate that the oxidation process is prerequisite for restructuring the Cu nanoparticles and enhancing their catalytic activity. These results also provide the direction

Table 2. Reaction rates (TOFs) for Cu catalysts before and after different treatments and corresponding TOF ratios (TOF_{After}/TOF_{Before}).^a

Entry	Catalyst	Reactant	Treatment condition (473 K, 0.25 h)	TOF _{Before} (h ⁻¹)	TOF _{After} (h ⁻¹)	TOF _{After} /TOF _{Before}
1	Cu/SiO ₂	ethanol	5 kPa ethanol + 5 kPa N ₂ O	15.0	60.2	4.0
2	Cu/SiO ₂	ethanol	5 kPa ethanol + 2 kPa O ₂	15.0	61.6	4.1
3	Cu/SiO ₂	ethanol	5 kPa ethanol + 5 kPa H ₂ O	15.0	18.3	1.2
4	Cu/SiO ₂	ethanol	5 kPa ethanol + 10 kPa Ar	15.0	15.3	1.0
5	Cu/SiO ₂	ethanol	5 kPa ethanol + 10 kPa H ₂	15.0	14.9	1.0
6	Cu/SiO ₂	ethanol	10 kPa H ₂ + 5 kPa N ₂ O	15.0	54.2	3.6
7	Cu/SiO ₂	ethanol	10 kPa Ar + 5 kPa N ₂ O	15.0	55.3	3.7
8	Cu/SiO ₂	ethanol	5 kPa ethanol + 5 kPa N ₂ O (453 K)	15.0	43.1	2.9
9	Cu/SiO ₂	ethanol	5 kPa ethanol + 5 kPa N ₂ O (463 K)	15.0	54.9	3.7
10	Cu/SiO ₂	ethanol	5 kPa ethanol + 5 kPa N ₂ O (493 K)	15.0	72.5	4.8
11 ^b	Cu/SiO ₂	1-propanol	3 kPa 1-propanol + 5 kPa N ₂ O	23.0	76.1	3.3
12 ^c	Cu/SiO ₂	2-propanol	3 kPa 2-propanol + 5 kPa N ₂ O	32.2	221.3	6.9
13 ^d	Cu/SiO ₂	2-butanol	3 kPa 2-butanol + 5 kPa N ₂ O	26.2	135.7	5.2
14 ^e	Cu/SiO ₂	acetone + H ₂	4 kPa acetone + 20 kPa H ₂ + 5 kPa N ₂ O	71.6	156.2	2.2
15 ^f	Cu/SiO ₂	methanol + H ₂ O	5 kPa methanol + 5 kPa H ₂ O + 5 kPa N ₂ O	24.6	56.1	2.3
16 ^g	Cu/SiO ₂ -IM	ethanol	5 kPa ethanol + 5 kPa N ₂ O	30.0	92.0	3.1
17 ^h	Cu/ZnO-CP	ethanol	5 kPa ethanol + 5 kPa N ₂ O	13.7	75.2	5.5

^aReaction conditions: 30 mL/min, 5 kPa ethanol, balance N₂, 473 K, ethanol conversion before treatment ~3.0%, selectivity to acetaldehyde > 95%.

^bReaction conditions: 30 mL/min, 3 kPa 1-propanol, balance N₂, 473 K, 1-propanol conversion before treatment ~3.0%, selectivity to propionaldehyde > 95%.

^cReaction conditions: 30 mL/min, 3 kPa 2-propanol, balance N₂, 473 K, 2-propanol conversion before treatment ~3.0%, selectivity to acetone > 95%.

^dReaction conditions: 30 mL/min, 3 kPa 2-butanol, balance N₂, 473 K, 2-butanol conversion before treatment ~3.0%, selectivity to 2-butanone > 95%.

^eReaction conditions: 30 mL/min, 4 kPa acetone, 20 kPa H₂, balance N₂, 473 K, acetone conversion before treatment ~5.0%, selectivity to 2-propanol > 92%.

^fReaction conditions: 30 mL/min, 5 kPa methanol, 5 kPa H₂O, balance N₂, 493 K, methanol conversion before treatment ~5.0%, selectivity to CO₂ > 98%.

^gCu/SiO₂-IM was prepared by incipient wetness impregnation method with 5.0% CuO loading and 16.7% Cu dispersion.

^hCu/ZnO-CP was prepared by coprecipitation method with 30.0% CuO loading and 5.7% Cu dispersion.

for engineering the Cu nanoparticles in a controllable way by tuning the treatment parameters. For example, use of N₂O, a mild oxidant, as discussed above, ensured the surface reconstruction of the Cu nanoparticles under the CH₃CH₂OH dehydrogenation conditions without detriment to their other structural factors (e.g. Cu particle size and dispersion).

3.5. Wide applicability of N₂O-involved treatment method. As a very important characteristic, the wide applicability of the N₂O treatment method was also examined for different Cu catalysts in different reactions (Entries 11–17, Table 2). Employing 1-propanol (Entry 11), 2-propanol (Entry 12), and 2-butanol (Entry 13), instead of ethanol, TOF_{After}/TOF_{Before} was 3.3, 6.9, and 5.2, respectively, in the dehydrogenation reactions to their respective aldehyde or ketones. Probed by a different type of reaction, acetone hydrogenation to 2-propanol, known as a less structure-sensitive reaction^{60–61}, Cu/SiO₂ after treatment still exhibited a

higher activity (156.2 h⁻¹ vs. 71.6 h⁻¹, TOF_{After}/TOF_{Before}=2.2) with almost unchanged selectivity (98.7% vs. 98.3%). Even for methanol steam reforming (to H₂ and CO₂), carried out under much harsher reaction conditions in the presence of H₂O, the N₂O treatment of Cu/SiO₂ also led to a much higher activity (56.1 h⁻¹ vs. 24.6 h⁻¹, TOF_{After}/TOF_{Before}=2.3) while retaining its CO₂ selectivity (98.7% vs. 98.3%). Cu/SiO₂ prepared differently, for example, by incipient wetness impregnation with a lower Cu loading (Entry 16), also showed an increase in its CH₃CH₂OH dehydrogenation activity by 3.1 folds after treatment. Cu/ZnO, the key catalyst component for methanol synthesis and other important reactions³³, exhibited a TOF_{After}/TOF_{Before} value of as high as 5.5 (Entry 17).

4. Conclusions

Clearly, the N₂O-involved treatment is facile and effective for controllably restructuring the Cu nanoparticles at

their surfaces, which lead to the formation of more coordinatively unsaturated Cu sites, and in turn to the dramatic enhancement in their catalytic reactivity for different reactions. The in situ TEM and XAS measurements provide direct evidence for visualizing the dynamic restructuring of the Cu nanoparticles under the reaction conditions and understanding the restructuring mechanism. Such controlled surface restructuring will be a viable approach, alternative to many sophisticated synthesis methods, to engineering the Cu catalysts and also potentially other metal catalysts (e.g. other Group IB metals) with high density of surface defects, catalytic efficiency and stability.

ASSOCIATED CONTENT

Supporting Information. This material is available free of charge via the Internet at <http://pubs.acs.org>. Detailed sample preparation and characterization methods, DFT calculations, **kinetic analysis**, and additional tables and figures.

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Notes

The authors declare no competing financial interest.

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REFERENCES

- (1) Schauermaun, S.; Nilius, N.; Shaikhutdinov, S.; Freund, H.-J., Nanoparticles for Heterogeneous Catalysis: New Mechanistic Insights. *Acc. Chem. Res.* **2013**, *46*, 1673-1681.
- (2) Campbell, C. T., The Energetics of Supported Metal Nanoparticles: Relationships to Sintering Rates and Catalytic Activity. *Acc. Chem. Res.* **2013**, *46*, 1712-1719.
- (3) Chng, L. L.; Erathodiyil, N.; Ying, J. Y., Nanostructured Catalysts for Organic Transformations. *Acc. Chem. Res.* **2013**, *46*, 1825-1837.
- (4) Liu, L.; Corma, A., Metal Catalysts for Heterogeneous Catalysis: From Single Atoms to Nanoclusters and Nanoparticles. *Chem. Rev.* **2018**, *118*, 4981-5079.
- (5) van den Berg, R.; Prieto, G.; Korpershoek, G.; van der Wal, L. I.; van Bunningen, A. J.; Lægsgaard-Jørgensen, S.; de Jongh, P. E.; de Jong, K. P., Structure Sensitivity of Cu and CuZn Catalysts Relevant to Industrial Methanol Synthesis. *Nat. Commun.* **2016**, *7*, 13057.
- (6) Liu, Y.; Zhao, G.; Wang, D.; Li, Y., Heterogeneous Catalysis for Green Chemistry based on Nanocrystals. *Natl. Sci. Rev.* **2015**, *2*, 150-166.
- (7) Zhang, Z.-C.; Xu, B.; Wang, X., Engineering Nanointerfaces for Nanocatalysis. *Chem. Soc. Rev.* **2014**, *43*, 7870-7886.
- (8) Munnik, P.; de Jongh, P. E.; de Jong, K. P., Recent Developments in the Synthesis of Supported Catalysts. *Chem. Rev.* **2015**, *115*, 6687-6718.
- (9) Tao, F.; Crozier, P. A., Atomic-Scale Observations of Catalyst Structures under Reaction Conditions and during Catalysis. *Chem. Rev.* **2016**, *116*, 3487-3539.
- (10) Jiang, Y.; Zhang, Z.; Yuan, W.; Zhang, X.; Wang, Y.; Zhang, Z., Recent Advances in Gas-Involved In Situ Studies via Transmission Electron Microscopy. *Nano Res.* **2018**, *11*, 42-67.
- (11) Kalz, K. F.; Kraehnert, R.; Dvoyashkin, M.; Dittmeyer, R.; Gläser, R.; Krewer, U.; Reuter, K.; Grunwaldt, J.-D., Future Challenges in Heterogeneous Catalysis: Understanding Catalysts under Dynamic Reaction Conditions. *ChemCatChem* **2017**, *9*, 17-29.
- (12) Schlögl, R., Heterogeneous Catalysis. *Angew. Chem. Int. Ed.* **2015**, *54*, 3465-3520.
- (13) Vendelbo, S. B.; Elkjær, C. F.; Falsig, H.; Puspitasari, I.; Dona, P.; Mele, L.; Morana, B.; Nelissen, B. J.; van Rijn, R.; Creemer, J. F.; Kooyman, P. J.; Helveg, S., Visualization of Oscillatory Behaviour of Pt Nanoparticles Catalysing CO Oxidation. *Nat. Mater.* **2014**, *13*, 884-890.
- (14) Li, Y.; Zakharov, D.; Zhao, S.; Tapper, R.; Jung, U.; Elsen, A.; Baumann, P.; Nuzzo, R. G.; Stach, E. A.; Frenkel, A. I., Complex Structural Dynamics of Nanocatalysts Revealed in Operando Conditions by Correlated Imaging and Spectroscopy Probes. *Nat. Commun.* **2015**, *6*, 7583.
- (15) Yoshida, H.; Kuwauchi, Y.; Jinschek, J. R.; Sun, K.; Tanaka, S.; Kohyama, M.; Shimada, S.; Haruta, M.; Takeda, S., Visualizing Gas Molecules Interacting with Supported Nanoparticle Catalysts at Reaction Conditions. *Science* **2012**, *335*, 317-319.
- (16) Ek, M.; Ramasse, Q. M.; Arnarson, L.; Georg Moses, P.; Helveg, S., Visualizing Atomic-Scale Redox Dynamics in Vanadium Oxide-Based Catalysts. *Nat. Commun.* **2017**, *8*, 305.
- (17) Neagu, D.; Papaioannou, E. I.; Ramli, W. K. W.; Miller, D. N.; Murdoch, B. J.; Ménéard, H.; Umar, A.; Barlow, A. J.; Cumpson, P. J.; Irvine, J. T. S.; Metcalfe, I. S., Demonstration of Chemistry at A Point through Restructuring and Catalytic Activation at Anchored Nanoparticles. *Nat. Commun.* **2017**, *8*, 1855.
- (18) Kamiuchi, N.; Sun, K.; Aso, R.; Tane, M.; Tamaoka, T.; Yoshida, H.; Takeda, S., Self-Activated Surface Dynamics in Gold Catalysts under Reaction Environments. *Nat. Commun.* **2018**, *9*, 2060.
- (19) Buurmans, I. L. C.; Weckhuysen, B. M., Heterogeneities of Individual Catalyst Particles in Space and Time as Monitored by Spectroscopy. *Nat. Chem.* **2012**, *4*, 873-886.
- (20) Aitbekova, A.; Wu, L.; Wrasman, C. J.; Boubnov, A.; Hoffman, A. S.; Goodman, E. D.; Bare, S. R.; Cargnello, M., Low-Temperature Restructuring of CeO₂-Supported Ru Nanoparticles Determines Selectivity in CO₂ Catalytic Reduction. *J. Am. Chem. Soc.* **2018**, *140*, 13736-13745.
- (21) Hernández Mejía, C.; van Deelen, T. W.; de Jong, K. P., Activity Enhancement of Cobalt Catalysts by Tuning Metal-Support Interactions. *Nat. Commun.* **2018**, *9*, 4459.
- (22) Dou, J.; Sun, Z.; Opalade, A. A.; Wang, N.; Fu, W.; Tao, F., Operando Chemistry of Catalyst Surfaces during Catalysis. *Chem. Soc. Rev.* **2017**, *46*, 2001-2027.
- (23) Vesborg, P. C. K.; Chorkendorff, I.; Knudsen, I.; Balmes, O.; Nerlov, J.; Molenbroek, A. M.; Clausen, B. S.; Helveg, S.,

Transient Behavior of Cu/ZnO-Based Methanol Synthesis Catalysts. *J. Catal.* **2009**, 262, 65-72.

(24) Freakley, S. J.; He, Q.; Harray, J. H.; Lu, L.; Crole, D. A.; Morgan, D. J.; Ntainjua, E. N.; Edwards, J. K.; Carley, A. F.; Borisevich, A. Y.; Kiely, C. J.; Hutchings, G. J., Palladium-Tin Catalysts for the Direct Synthesis of H₂O₂ with High Selectivity. *Science* **2016**, 351, 965-968.

(25) Eren, B.; Zherebetskyy, D.; Patera, L. L.; Wu, C. H.; Bluhm, H.; Africh, C.; Wang, L.-W.; Somorjai, G. A.; Salmeron, M., Activation of Cu (111) Surface by Decomposition into Nanoclusters Driven by CO Adsorption. *Science* **2016**, 351, 475-478.

(26) Lee, A. F.; Wilson, K., Structure-Reactivity Correlations in the Selective Aerobic Oxidation of Cinnamyl Alcohol: In Situ XAFS. *Green Chem.* **2004**, 6, 37-42.

(27) Chenna, S.; Banerjee, R.; Crozier, P. A., Atomic-Scale Observation of the Ni Activation Process for Partial Oxidation of Methane Using In Situ Environmental TEM. *ChemCatChem* **2011**, 3, 1051-1059.

(28) Sushkevich, V. L.; Ivanova, I. I.; Taarning, E., Mechanistic Study of Ethanol Dehydrogenation over Silica-Supported Silver. *ChemCatChem* **2013**, 5, 2367-2373.

(29) Wang, Z.; Xu, Y.; El-Soda, M.; Lucci, F. R.; Madix, R. J.; Friend, C. M.; Sykes, E. C. H., Surface Structure Dependence of the Dry Dehydrogenation of Alcohols on Cu(111) and Cu(110). *J. Phys. Chem. C* **2017**, 121, 12800-12806.

(30) Graciani, J.; Mudiyanse, K.; Xu, F.; Baber, A. E.; Evans, J.; Senanayake, S. D.; Stacchiola, D. J.; Liu, P.; Hrbek, J.; Sanz, J. F.; Rodriguez, J. A., Highly Active Copper-Ceria and Copper-Ceria-Titania Catalysts for Methanol Synthesis from CO₂. *Science* **2014**, 345, 546-550.

(31) Behrens, M.; Studt, F.; Kasatkin, I.; Köhl, S.; Hävecker, M.; Abild-Pedersen, F.; Zander, S.; Girgsdies, F.; Kurr, P.; Knief, B.-L.; Tovar, M.; Fischer, R. W.; Nørskov, J. K.; Schlögl, R., The Active Site of Methanol Synthesis over Cu/ZnO/Al₂O₃ Industrial Catalysts. *Science* **2012**, 336, 893-897.

(32) Kuld, S.; Thorhauge, M.; Falsig, H.; Elkjær, C. F.; Helveg, S.; Chorkendorff, I.; Sehested, J., Quantifying the Promotion of Cu Catalysts by ZnO for Methanol Synthesis. *Science* **2016**, 352, 969-974.

(33) Kondrat, S. A.; Smith, P. J.; Wells, P. P.; Chater, P. A.; Carter, J. H.; Morgan, D. J.; Fiordaliso, E. M.; Wagner, J. B.; Davies, T. E.; Lu, L.; Bartley, J. K.; Taylor, S. H.; Spencer, M. S.; Kiely, C. J.; Kelly, G. J.; Park, C. W.; Rosseinsky, M. J.; Hutchings, G. J., Stable Amorphous Georgeite as A Precursor to A High-Activity Catalyst. *Nature* **2016**, 531, 83-87.

(34) Palo, D. R.; Dagle, R. A.; Holladay, J. D., Methanol Steam Reforming for Hydrogen Production. *Chem. Rev.* **2007**, 107, 3992-4021.

(35) Bolder, F. H. A., Dehydrogenation of Alcohol Mixtures to Esters and Ketones. *Ind. Eng. Chem. Res.* **2008**, 47, 7496-7500.

(36) Jin, M.; He, G.; Zhang, H.; Zeng, J.; Xie, Z.; Xia, Y., Shape-Controlled Synthesis of Copper Nanocrystals in an Aqueous Solution with Glucose as a Reducing Agent and Hexadecylamine as a Capping Agent. *Angew. Chem. Int. Ed.* **2011**, 50, 10560-10564.

(37) Mott, D.; Galkowski, J.; Wang, L.; Luo, J.; Zhong, C.-J., Synthesis of Size-Controlled and Shaped Copper Nanoparticles. *Langmuir* **2007**, 23, 5740-5745.

(38) Creemer, J. F.; Helveg, S.; Hoveling, G. H.; Ullmann, S.; Molenbroek, A. M.; Sarro, P. M.; Zandbergen, H. W., Atomic-Scale Electron Microscopy at Ambient Pressure. *Ultramicroscopy* **2008**, 108, 993-998.

(39) Jiang, Y.; Li, H.; Wu, Z.; Ye, W.; Zhang, H.; Wang, Y.; Sun, C.; Zhang, Z., In Situ Observation of Hydrogen-Induced Surface

Faceting for Palladium-Copper Nanocrystals at Atmospheric Pressure. *Angew. Chem.* **2016**, 128, 12615-12618.

(40) Xu, M.; Yao, S.; Rao, D.; Niu, Y.; Liu, N.; Peng, M.; Zhai, P.; Man, Y.; Zheng, L.; Wang, B.; Zhang, B.; Ma, D.; Wei, M., Insights into Interfacial Synergistic Catalysis over Ni@TiO_{2-x} Catalyst toward Water-Gas Shift Reaction. *J. Am. Chem. Soc.* **2018**, 140, 11241-11251.

(41) Jinschek, J. R., Advances in the Environmental Transmission Electron Microscope (ETEM) for Nanoscale In Situ Studies of Gas-Solid Interactions. *Chem. Commun.* **2014**, 50, 2696-2706.

(42) Coq, B.; Figueras, F., Structure-Activity Relationships in Catalysis by Metals: Some Aspects of Particle Size, Bimetallic and Supports Effects. *Coord. Chem. Rev.* **1998**, 178-180, 1753-1783.

(43) Liu, H.; Huang, Z.; Xia, C.; Jia, Y.; Chen, J.; Liu, H., Selective Hydrogenolysis of Xylitol to Ethylene Glycol and Propylene Glycol over Silica Dispersed Copper Catalysts Prepared by a Precipitation-Gel Method. *ChemCatChem* **2014**, 6, 2918-2928.

(44) Ravel, B.; Newville, M., ATHENA, ARTEMIS, HEPHAESTUS: Data Analysis for X-ray Absorption Spectroscopy Using IFEFFIT. *J. Synchrotron Radiat.* **2005**, 12, 537-541.

(45) Alayon, E. M. C.; Nachtegaal, M.; Kleymanov, E.; van Bokhoven, J. A., Determination of the Electronic and Geometric Structure of Cu Sites during Methane Conversion over Cu-MOR with X-Ray Absorption Spectroscopy. *Microporous Mesoporous Mater.* **2013**, 166, 131-136.

(46) Kresse, G.; Furthmüller, J., Efficiency of Ab-Initio Total Energy Calculations for Metals and Semiconductors Using A Plane-Wave Basis Set. *Comput. Mater. Sci.* **1996**, 6, 15-50.

(47) Kresse, G.; Furthmüller, J., Efficient Iterative Schemes for Ab Initio Total-Energy Calculations Using A Plane-Wave Basis Set. *Phys. Rev. B* **1996**, 54, 11169-11186.

(48) Blöchl, P. E., Projector Augmented-Wave Method. *Phys. Rev. B* **1994**, 50, 17953-17979.

(49) Kresse, G.; Joubert, D., From ultrasoft pseudopotentials to the projector augmented-wave method. *Phys. Rev. B* **1999**, 59, 1758-1775.

(50) Perdew, J. P.; Burke, K.; Ernzerhof, M., Generalized Gradient Approximation Made Simple. *Phys. Rev. Lett.* **1996**, 77, 3865-3868.

(51) Grimme, S.; Ehrlich, S.; Goerigk, L., Effect of the Damping Function in Dispersion Corrected Density Functional Theory. *J. Comput. Chem.* **2011**, 32, 1456-1465.

(52) Henkelman, G.; Jónsson, H., A Dimer Method for Finding Saddle Points on High Dimensional Potential Surfaces Using Only First Derivatives. *J. Chem. Phys.* **1999**, 111, 7010-7022.

(53) Henkelman, G.; Uberuaga, B. P.; Jónsson, H., A Climbing Image Nudged Elastic Band Method for Finding Saddle Points and Minimum Energy Paths. *J. Chem. Phys.* **2000**, 113, 9901-9904.

(54) Liu, H.; Huang, Z.; Han, Z.; Ding, K.; Liu, H.; Xia, C.; Chen, J., Efficient Production of Methanol and Diols via the Hydrogenation of Cyclic Carbonates Using Copper-Silica Nanocomposite Catalysts. *Green Chem.* **2015**, 17, 4281-4290.

(55) Sad, M. E.; Neurock, M.; Iglesia, E., Formation of C-C and C-O Bonds and Oxygen Removal in Reactions of Alkanediols, Alkanols, and Alkanals on Copper Catalysts. *J. Am. Chem. Soc.* **2011**, 133, 20384-20398.

(56) Sziyalski, A.; Girgsdies, F.; Rabis, A.; Wang, Y.; Niederberger, M.; Ressler, T., In Situ Investigations of Structure-Activity Relationships of A Cu/ZrO₂ Catalyst for the Steam Reforming of Methanol. *J. Catal.* **2005**, 233, 297-307.

(57) Hansen, P. L.; Wagner, J. B.; Helveg, S.; Rostrup-Nielsen, J. R.; Clausen, B. S.; Topsøe, H., Atom-Resolved Imaging of

Dynamic Shape Changes in Supported Copper Nanocrystals. *Science* **2002**, 295, 2053-2055.

(58) Nolte, P.; Stierle, A.; Kasper, N.; Jin-Phillipp, N. Y.; Jeutter, N.; Dosch, H., Reversible Shape Changes of Pd Nanoparticles on MgO(100). *Nano Lett.* **2011**, 11, 4697-4700.

(59) Avanesian, T.; Dai, S.; Kale, M. J.; Graham, G. W.; Pan, X.; Christopher, P., Quantitative and Atomic-Scale View of CO-Induced Pt Nanoparticle Surface Reconstruction at Saturation Coverage via DFT Calculations Coupled with in Situ TEM and IR. *J. Am. Chem. Soc.* **2017**, 139, 4551-4558.

(60) Fuente, A. M.; Pulgar, G.; González, F.; Pesquera, C.; Blanco, C., Activated Carbon Supported Pt Catalysts: Effect of

Support Texture and Metal Precursor on Activity of Acetone Hydrogenation. *Appl. Catal. A* **2001**, 208, 35-46.

(61) Sen, B.; Vannice, M. A., Metal-Support Effects on Acetone Hydrogenation over Platinum Catalysts. *J. Catal.* **1988**, 113, 52-71.

Table of Contents/Abstract Graphics

